Xu (Larry) Duan

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EDUCATION

The University of Texas at Austin Aug. 2024 - May 2026 (Expected)
Master of Science in Mechanical Engineering, Dynamic System and Control Area Austin, TX

- Co-Advised by Dr. van Oort and Dr. Chen
- Relevant Coursework: Engineering Analysis, Modeling of Physical System, Transport Phenomenon, Dynamics of Mechanical System.

Shanghai Jiao Tong University Sep. 2019 - Jul. 2023
Bachelor of Engineering in Naval Architecture and Marine Engineering Shanghai, CN

- GPA: **3.79/4.3, 89/100**
- Advised by Dr. Fu
- Relevant Coursework: Finite Element Method, Computational Fluid Dynamics, Modern Control, Matrix Theory, Digital Control, Marine Structural Mechanics, Marine Hydrodynamics, Convex Optimization, Advanced Thermodynamics, Thinking and Approach of Programming (C++), Data Structures (C++)

HONORS & AWARDS

Philip C. and Linda L. Lewis Foundation Graduate Fellowship Aug. 2024
CCS (China ship Classification Society) Scholarship (10%) Sep. 2022 & Sep. 2020
Shengshen Scholarship (5%) Sep. 2020

INDUSTRIAL EXPERIENCE

RAPID Aug. 2024 - Present
Well Hydraulics and Thermodynamics Research Assistant, Supervisor: Dr. van Oort Austin, TX

- Developed a two-phase fluid model to analyze well hydraulics and thermodynamics, using the Finite Volume Method (FVM) to predict pressure profile, flow rates, and temperature with less than 5% relative error compared to experimental data.
- Implemented a control model for bottomhole pressure and temperature, effectively mitigating blowout risks.
- Conducted Fluid-Structure Interaction (FSI) analysis using the Lattice Boltzmann Method (LBM) and validated the results against Ansys simulations, achieving a difference of less than 3%.

State Key Laboratory of Ocean Engineering Sep. 2023 - Jun. 2024
Ocean Engineering Research Assistant, Supervisor: Dr. Fu Shanghai, CN

- Wrote a Fortran program to numerically solve fully nonlinear Boussinesq Wave Model from scratch.
- Applied the Monotonic Upstream-centered Scheme for Conservation Law (MUSCL) for spatial discretizing and utilized the third-order Strong Stability-Preserving (SSP) Runge-Kutta scheme for time integration.

Haipeng Marine Engineering (Shanghai) Co. Jul. 2023 - Aug. 2023
Hydrodynamics and Structural Engineering Intern Shanghai, CN

- Calculated the hydrodynamic coefficients in HydroStar and Morison loads of the fish farming nets and cages.
- Analyzed the responses of large net cages under regular wave excitation in time domain using the coupled rigid-body motion equation in Orcaflex and conducted structural analysis in Abaqus, showing that the average increase of net tension at key positions is 51% when considering the influence of floating bodies on the net.
- Conducted hydro-elastic analysis of fish farming vessels using the coupled hydroelastic equation and found that the bending moment of the elastic model at the section near the middle is 48% that of the rigid model.

ACADEMIC EXPERIENCE

Shanghai Jiao Tong University Jul. 2023 - Aug. 2023
Teaching Assistant for Junior-level Structural Dynamics Shanghai, CN

- Guided junior students in learning Abaqus and constructing a basic ship model to do modal analysis.
- Taught students how to apply the FEM to develop a program to solve non-homogeneous beam bending problems.

SELECTED PROJECTS

Shallow Water Wave Simulation using (FVM)

- Simulated water wave propagation over shallow, variable-depth terrain using the Finite Volume Method (FVM) in FUNWAVE, developed in Fortran on a Linux system.
- Applied higher-order numerical techniques to discretize fully nonlinear Boussinesq wave equations for accuracy.
- Developed a Python program to dynamically visualize the time evolution of waves.

Control System Design for an Underwater Vehicle

- Developed a mathematical model and simulation of a 6-DOF underwater vehicle with voltage distribution for different propellers in Simulink.

- Designed a Proportional Integral Differential (PID) control system to correct deviations between the robot's actual depth and heading and target depth and heading with overshoot 18.50% and 31.57% and settling time 5.13s and 12.84s for depth and heading, respectively.

Thermal and Mechanical Simulation of a CPU Heatsink with SolidWorks

- Built a detailed 3D model in SolidWorks to accurately capture the complex geometry of the CPU heatsink.
- Performed thermal and mechanical simulations in SolidWorks to evaluate stress distribution and temperature behavior across the heatsink under operational conditions.

Solving Linear Advection Equation using Streamline-Upwind Petrov-Galerkin (SUPG) Method

- Solved the varying coefficient linear advection equation with adaptive mesh refinement using deal.II on a Linux system.
- Utilized parallel computing with multiple processors accessing shared memory techniques in deal.II to assemble the global matrix and accelerate computation.

Comparison between Finite Element, Finite Volume and Finite Difference solutions of Lid-driven Cavity Flow

- Solved the Lid-driven Cavity Flow problem under different Reynolds numbers using finite difference method, SIMPLE algorithm with finite volume discretization, and Galerkin finite element method, respectively.
- Finite difference method is shown to be fastest and most robust to changes in flow, but is not as accurate as the SIMPLE implementation.

Book Lamp Control System using PLC

- Designed and implemented a control system for opening and closing book lamps using Programmable Logic Controllers (PLC).
- Developed and debugged a program to process analog signals from a photosensitive resistor, converting them into digital signals to monitor and adjust light intensity.

Wind Speed Prediction with Neural Networks

- Conducted short-term wind speed forecasting using a Long Short-Term Memory (LSTM) neural network based on historical wind speed data.
- Employed signal processing techniques to preprocess raw wind speed data for improved model accuracy.

PUBLICATIONS

- X. Duan. Fluid-Structure Interaction Analysis of Floating Structures under Shallow Water Waves. *Shanghai Jiao Tong University Bachelor Dissertation*, Shanghai, China, Jun 2023.
- X. Duan, B. Song, and S. Fu. Hydroelastic analysis of fish farming vessels under using Boussinesq Equations. *35th China National Hydrodynamics Symposium*, Harbin, China, Dec 2023.

Working:

- X. Duan, Y. Zhang, D. Chen, and E. van Oort. A Higher Precision Real-time Controller Based on System Identification for Managed Temperature Drilling.
- X. Duan, Y. Zhang, D. Chen, and E. van Oort. Coupled Managed Pressure and Managed Temperature Drilling Based on a MIMO controller in Geothermal and HPHT Wells.

SKILLS AND QUALIFICATIONS

Programming: Python, C/C++, Linux, MATLAB, Fortran (with MPI & OpenMP)
Software: SolidWorks, Ansys, Abaqus, Paraview, deal.II, PETSc, AutoCad, OrcaFlex, HydroStar, Office suite
IELTS: 7.0 (Listening: 7.5, Reading 8.0, Writing: 7.0, Speaking: 6.0)
GRE: 336+4.0 (Verbal: 166, Quantitative: 170, Analytical Writing: 4.0)

Transferable Attributes:

- Superior diagnostic skills for issues and innovative invention practices;
- Capacity to coordinate and optimize solutions and conduct flexible commissioning;
- Collaborative teamwork skills;
- Foresight and vision to evaluate the complexity of projects.